

Ishika Khapekar

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EDUCATION

B.Tech — Computer Science & Engineering

2022 – 2026

G H Raisonni University, Nagpur | CGPA: 8.3 / 10

Class XII — Prerna Junior College, Nagpur

2020 – 2022

Percentage: 88%

EXPERIENCE

Data Analyst Intern — Cojag Smart Technology, Nagpur

June 2025 – December 2025

- Cleaned and structured 5,000+ records maintaining high data accuracy for internal reporting — directly supporting data consolidation and cleaning workflows critical to business operations.
 - Performed quantitative analysis identifying operational inefficiencies across departments — presented findings via Power BI dashboards to management stakeholders at multiple reporting levels.
 - Built interactive data visualisation UI supporting real-time decision-making and billing query resolution for cross-functional engineering teams.
 - Collaborated proactively with cross-functional teams aligning analytical outputs with business objectives demonstrating flexibility across simultaneous workstreams.
 - Worked with annotating 2000+ images for detection.
 - Did further used these annotated images for detection by modelling it first and then using it for detection. Project contributed were Licence Plate Detection and Weed Detection.
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PROJECTS

Crop Yield Forecasting & Food Security Risk Assessment

- Built end-to-end ML forecasting pipeline on 19,000+ agricultural records (1997–2020) using IMD rainfall, fertiliser, and crop data across all Indian states.
- Trained and benchmarked Linear Regression, Random Forest, and XGBoost models; XGBoost achieved highest R^2 with lowest RMSE — SHAP values identified fertiliser as top yield driver.
- Classified all states into Surplus, Near-Deficit, and Deficit food security zones with Folium choropleth visualisation.

GitHub: github.com/IshikaKhapekar/crop-yield-forecasting

Amazon Prime — Subscription & Streaming Behaviour Analytics

- Analysed 60,000+ subscriber records across demographics, viewing patterns, and plan types — performed RFM segmentation to identify high-engagement vs. churn-risk subscriber cohorts.
- Built Logistic Regression and Random Forest models to predict subscription churn with 87% accuracy — identified watch frequency and content genre as top churn predictors using feature importance.
- Designed Power BI dashboard tracking active subscribers, churn rate, average revenue per user, and content engagement trends — delivered actionable insights for retention strategy.

GitHub: github.com/IshikaKhapekar/amazon-prime-analytics

India Unemployment Pattern Analysis & Scheme Effectiveness Scoring

- Analysed 740 records of state-wise monthly unemployment (2019–2020) across rural and urban areas using PLFS data; quantified COVID-19 impact via 2019 vs. 2020 comparative analysis.
- Built Random Forest and XGBoost classifiers to predict unemployment risk category (Low/Medium/High); applied K-Means clustering with PCA validation to profile states into tiers.
- Ran NLP keyword analysis on MGNREGS, PMKVY, PM SVANidhi, and PMEGP scheme descriptions — built scheme coverage heatmap mapping which schemes target which unemployed groups.

GitHub: github.com/IshikaKhapekar/unemployment-analysis-india

India Public Health Infrastructure & Disease Burden Analysis

- Built composite Health Score (0–100) for 36 Indian states using 14 NFHS-5 indicators; normalised positive and negative indicators to enable cross-state comparison.
- Performed K-Means clustering with PCA validation — classified all states into Strong, Moderate, Developing, and Critical health tiers with interactive Folium map.
- Applied SHAP analysis on trained XGBoost classifier; identified sanitation, institutional delivery, and immunisation as top predictors of health tier.

GitHub: github.com/lshikaKhapekar/health-infrastructure-india

India School Education Quality & Dropout Pattern Analysis

- Analysed state-wise school dropout rates by gender and education level using DISE and UDISE datasets — built Education Quality Index from 11 indicators.
- Identified female literacy as strongest predictor of dropout reduction; confirmed girls toilet coverage directly correlates with girls' secondary retention across all states.
- Applied NLP keyword analysis on NEP 2020 policy language to map goals against data gaps, and clustered states into education tiers using K-Means.

GitHub: github.com/lshikaKhapekar/education-dropout-india

TECHNICAL SKILLS

Languages	Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost), SQL, SQLite
ML & DS	Supervised Learning (Regression, Classification), Unsupervised Learning (K-Means, PCA), NLP (NLTK, TF-IDF), Time Series, SHAP, Feature Engineering
Visualisation	Power BI (DAX, Drill-through, Slicers, Gauge, Matrix), Matplotlib, Seaborn, Plotly, Folium
Data Tools	MS Excel (VLOOKUP, XLOOKUP, Power Query, Pivot Tables), Google Colab, Jupyter Notebook
Dev Tools	Git, GitHub, Flask, VS Code, Figma
Concepts	EDA, Composite Index Design, Dashboard Reporting, Data Cleaning, Stakeholder Communication, Policy Data Analysis, Leadership

CERTIFICATIONS

Python for Data Science, AI & Development — IBM, Coursera [view](#)

Databases and SQL for Data Science with Python — IBM, Coursera [view](#)

Excel Essentials for Data Analytics — IBM, Coursera [view](#)

EXTRACURRICULAR ACTIVITIES

Secretary — Student Council, G H Raison University

- Led coordination of 10+ academic and cultural events managing cross-functional student teams of 80+ members — developed stakeholder management and organisational skills applicable to fast-paced environments.